WORKED SOLUTION

STEP II 1991, Question 8

Pure • Nonlinear First-Order ODEs

STEP II 1991, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q8

The key idea

Both differential equations are quadratics in $\frac{\mathrm{d}y}{\mathrm{d}x}$, so we can solve for the derivative algebraically first and only then integrate. Each quadratic has two roots, so we get two candidate first-order equations, and the conditions at $x = 0$ decide which branch is allowed. Hyperbolic identities, in particular $\cosh^2 x - \sinh^2 x = 1$ and the half-angle formulae, keep the roots tidy.

Solution

The quadratic

By the quadratic formula,

$$u = \frac{-2\sinh x \pm \sqrt{4\sinh^2 x + 4}}{2} = -\sinh x \pm \sqrt{\sinh^2 x + 1} = -\sinh x \pm \cosh x,$$

using $\cosh^2 x - \sinh^2 x = 1$ and $\cosh x > 0$. Since $\cosh x - \sinh x = \mathrm{e}^{-x}$ and $\cosh x + \sinh x = \mathrm{e}^{x}$, the two roots are

$$u = \mathrm{e}^{-x} \qquad \text{or} \qquad u = -\mathrm{e}^{x}.$$

(Check: their product is -1 and their sum is $\mathrm{e}^{-x} - \mathrm{e}^{x} = -2\sinh x$, as they should be.)

The first differential equation

With $u = \frac{\mathrm{d}y}{\mathrm{d}x}$ this is exactly the quadratic above, so at each x either $y' = \mathrm{e}^{-x}$ or $y' = -\mathrm{e}^{x}$.

Ask yourself Could y' follow one root for some values of x and the other root elsewhere?

No. The first root is always positive and the second always negative, so they never meet. A derivative has the intermediate value property (Darboux's theorem; and if

you assume y' is continuous, the ordinary intermediate value theorem), so y' can never jump from a positive value to a negative one without passing through 0, which neither root takes. Hence y' stays on one root throughout.

At $x = 0$ the roots are 1 and -1, and we need $y' > 0$, so $\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{-x}$. Integrating, $y = -\mathrm{e}^{-x} + c$, and $y = 0$ at $x = 0$ gives $c = 1$:

$$\boxed{y = 1 - \mathrm{e}^{-x}}$$

The second differential equation

Write $p = \frac{\mathrm{d}y}{\mathrm{d}x}$, so $p^2 \sinh x + 2p - \sinh x = 0$.

At $x = 0$, $\sinh 0 = 0$ and the equation reduces to $2p = 0$, so we must have $y'(0) = 0$.

For $x \neq 0$, $\sinh x \neq 0$ and the quadratic formula gives

$$p = \frac{-2 \pm \sqrt{4 + 4\sinh^2 x}}{2\sinh x} = \frac{-1 \pm \cosh x}{\sinh x}.$$

Using $\cosh x = 1 + 2\sinh^2 \frac{x}{2} = 2\cosh^2 \frac{x}{2} - 1$ and $\sinh x = 2\sinh \frac{x}{2} \cosh \frac{x}{2}$, the two roots simplify to

$$\frac{\cosh x - 1}{\sinh x} = \frac{2\sinh^2 \frac{x}{2}}{2\sinh \frac{x}{2}\cosh \frac{x}{2}} = \tanh \tfrac{x}{2},$$
$$\frac{-1 - \cosh x}{\sinh x} = -\frac{2\cosh^2 \frac{x}{2}}{2\sinh \frac{x}{2}\cosh \frac{x}{2}} = -\coth \tfrac{x}{2}.$$

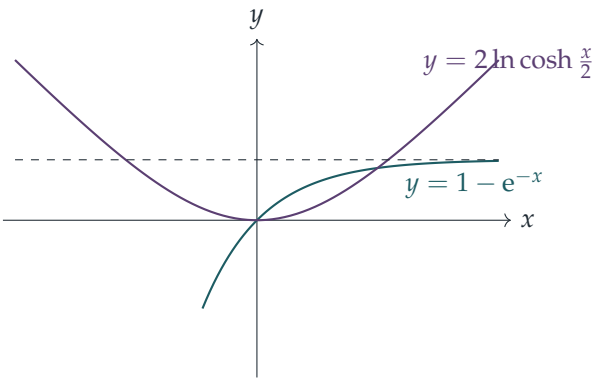
Choosing the branch. The second root $-\coth \frac{x}{2}$ is unbounded as $x \to 0$, and integrating it gives $y = -2\ln|\sinh \frac{x}{2}| + c$, which tends to $+\infty$ as $x \to 0$ from either side. So on neither side of the origin can this branch give a solution that is continuous at $x = 0$ with $y(0) = 0$. Near $x = 0$ (on each side) we must therefore take the first root, which agrees with $y'(0) = 0$ since $\tanh 0 = 0$. The two roots never coincide (one has the sign of x, the other the opposite sign, and neither vanishes for $x \neq 0$), so by the same intermediate value argument as before the solution stays on this branch for all x:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tanh \tfrac{x}{2} \quad \text{for all } x.$$

Since $\frac{\mathrm{d}}{\mathrm{d}x} \cosh \frac{x}{2} = \frac{1}{2}\sinh \frac{x}{2}$, we have $\tanh \frac{x}{2} = 2 \cdot \frac{\frac{1}{2}\sinh \frac{x}{2}}{\cosh \frac{x}{2}}$, which is of the form $2f'/f$, so integrating gives $y = 2\ln \cosh \frac{x}{2} + c$ (no modulus is needed as $\cosh > 0$). The condition $y(0) = 0$ gives $c = 0$, so

$$\boxed{y = 2\ln \cosh \tfrac{x}{2} = \ln\left(\frac{1 + \cosh x}{2}\right)}$$

the second form following from $2\cosh^2 \frac{x}{2} = 1 + \cosh x$.



Checks

- For $y = 1 - \mathrm{e}^{-x}$: $y'^2 + 2y' \sinh x = \mathrm{e}^{-2x} + \mathrm{e}^{-x}(\mathrm{e}^{x} - \mathrm{e}^{-x}) = 1$, as required, and $y'(0) = 1 > 0$.
- For $y = 2\ln\cosh\frac{x}{2}$, write $t = \tanh\frac{x}{2}$, so $\sinh x = \frac{2t}{1-t^2}$; then $\sinh x\,(t^2 - 1) + 2t = -2t + 2t = 0$. The solution is even, as it should be, because the equation is unchanged by $x \mapsto -x$, $y \mapsto y$.
- Numerically, at $x = 1$ the solutions are $1 - \mathrm{e}^{-1} \approx 0.6321$ and $2\ln\cosh\frac{1}{2} \approx 0.2402$; substituting finite-difference derivatives at many random points satisfies both equations to within 10^{-6}.

Common slips

- Dividing the second equation by $\sinh x$ without noticing that it vanishes at $x = 0$, which is exactly where the initial condition lives; the original equation forces $y'(0) = 0$.
- Keeping both branches in the second part: the $-\coth\frac{x}{2}$ branch cannot satisfy $y(0) = 0$, because its integral blows up at the origin.

Beyond STEP

Equations that are not linear in y', such as $F(x, y, y') = 0$, can have several solutions through one point, or "singular" solutions that no single branch describes, so existence and uniqueness theorems (Picard–Lindelöf) are stated for equations already solved in the form $y' = f(x, y)$. Here the roots were separated, so each branch behaved well; when the discriminant can vanish, branches can be glued together and uniqueness fails. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence, uniqueness and these nonlinear first-order equations.

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